Towards a Light-Weight Segmentation Model for Martian Navigation

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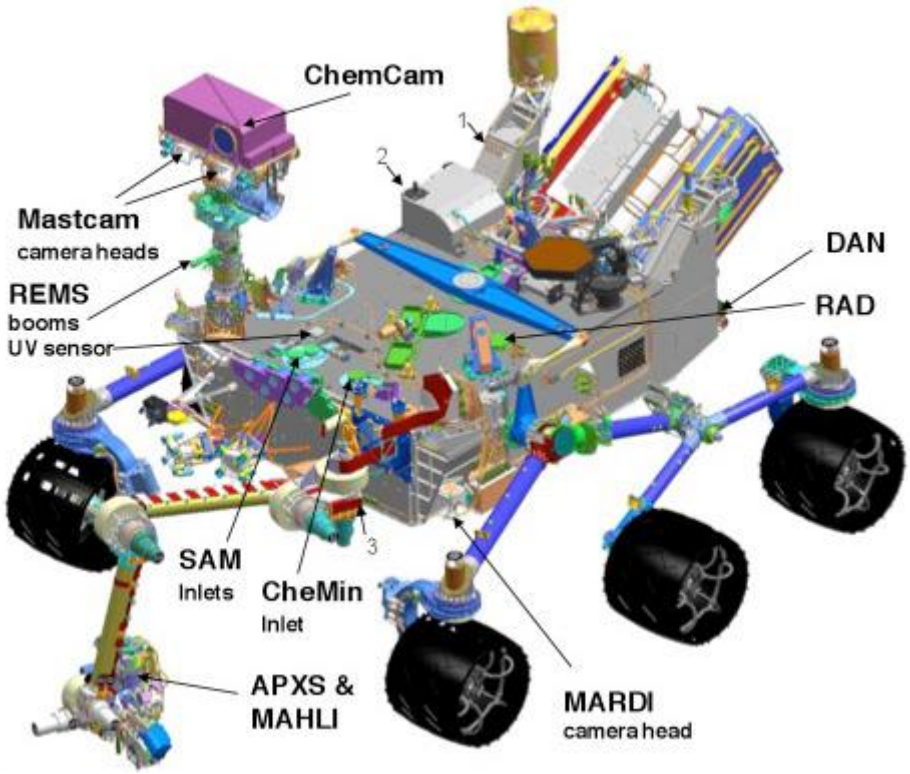
Overview

This project explores the effects of reducing the parameter size for a UNet model's performance on semantic segmentation of Mars Rover Curiosity's image data for the purposes of navigation. We will train two models, a full-sized UNet model and one with significantly reduced parameters. We'll summarize the results, examine ethical and bias concerns before discussing possible future work.

Figure 1 Curiosity Rover



Dataset and Task

Our task will be to look at the difference in performance in the UNet model (Ronneberger 2015) on Curiosity Rover's navigational camera when we reduce the number of parameters. This would allow the model to potentially use less power, run faster, and require less storage space

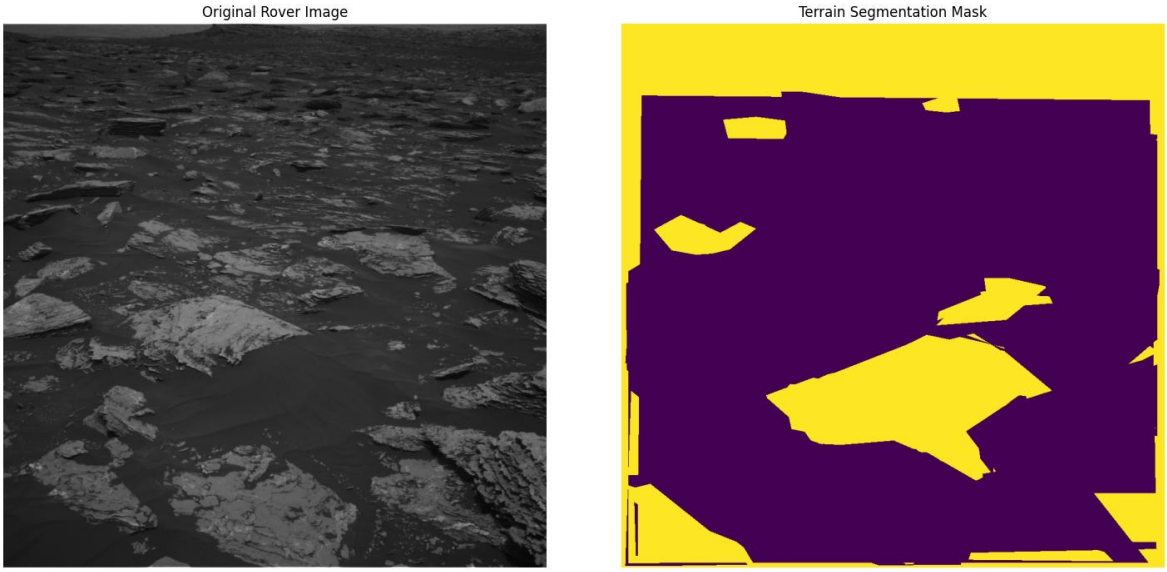
(Yang 2023). To support this experiment, we used the AI4Mars dataset (Swan 2021). We extracted the images and ground truth labels related to the Navigation Mast on the Rover. The ground truth labels provided by the AI4Mars dataset come in four categories:

- Soil
- Bedrock
- Sand
- Big Rock

As we are interested in building a model for navigation, we remapped these categories into a binary classification problem: Soil/Bedrock are marked as navigable. Sand/Big Rock are marked as non-navigable. Any pixels which were left unlabeled are also marked as non-navigable.

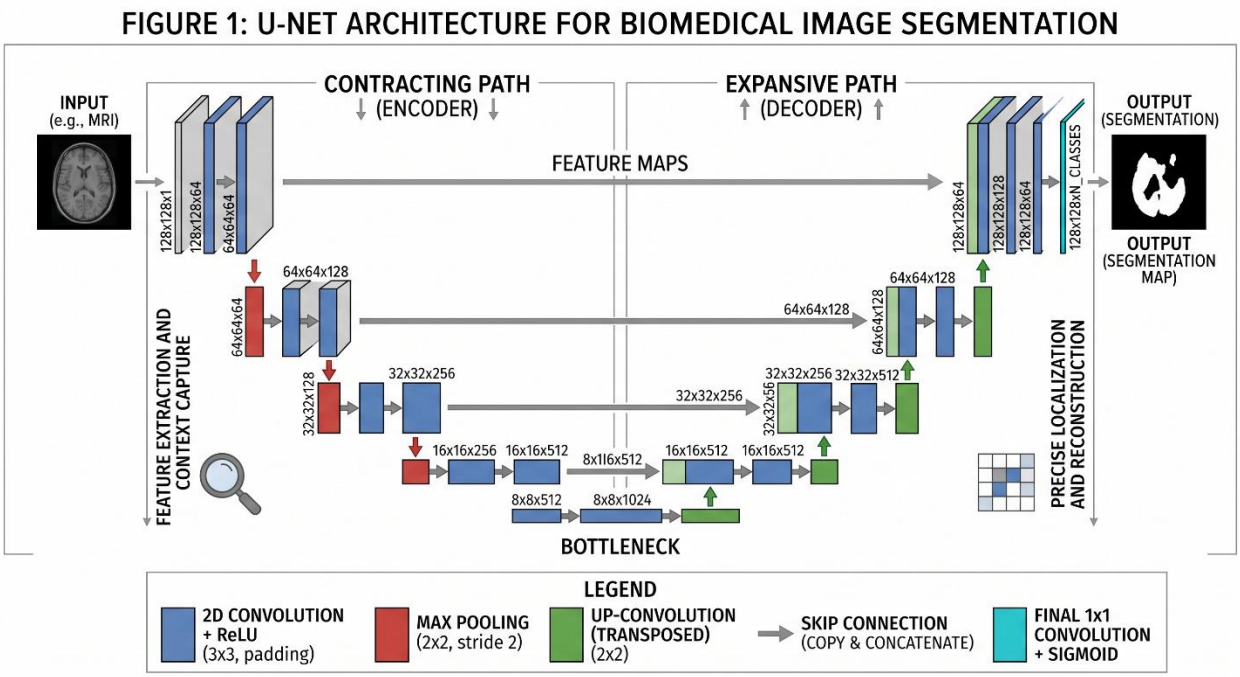
After extracting the Curiosity images from the dataset, we ended up with 16064 image/mask pairs for training our models.

Figure 2 Sample Image from AI4Mars Dataset



Model Architecture and Design

Figure 3 UNet Architecture - Generated with AI Assistance



The baseline model we built was based on UNet – a model originally intended for solving semantic segmentation problems in the medical industry. A diagram of this model can be seen in figure N. We deviated slightly from the original implementation by adding a BatchNorm operation to the convolution module. This model had 15,329,473 parameters.

UNet works by having two parallel encoders bridged by a convolution layer. Each encoder has a skip connection to the encoder of a similar size on the opposite end of the network. Encoders on the input side of the network effectively down samples the image via MaxPool, with each step having an increasing number of filters. Each step on the output side of the network Up samples the image via Inverse Convolutions.

Our comparison model used the same architecture as the UNet architecture. The primary change for the comparison model, which we'll refer to as Mini-UNet, is the reduction of the number of filters for each step in the convolution modules by 50%. This version of the model has 960,049 parameters.

Experimental Comparison

On autonomous devices like the Mars Rover power and space can be at a premium. While semantic segmentation could be helpful for navigation, large segmentation models could be power hungry. We perform an experiment to see the effects of reducing the number of parameters in a UNet to see if the smaller version can keep up.

We split the Curiosity training data into three distinct sets, keeping 70% for training, 15% for validation; and 15% for testing. Before being fed into the model, each image underwent the following transformations:

- Convert to gray scale.

- Reduce size to 512x512. Note that when resizing the training masks we used a Nearest Neighbor algorithm to avoid corrupting the label data.
- Standardized the inputs.

Dataset	**Size**
Training	11244
Validation	2409
Test	2411

The same train-validation-test split was kept constant when training both the baseline and comparison models.

Each model was trained using an Adam Optimizer, a learning rate of 3e-4, the Dice Loss function. Training was done for 5 epochs for each model.

Each training session used the same loss function – Dice Loss (Jadon, S 2020). This loss function measures the similarity between two images. In this case we use it to measure how similar the model's output image is compared to the ground truth mask.

$$DL(y,p) = 1 - \frac{(2yp + 1)}{y + p + 1}$$

Here "y" is the ground truth and "p" is the predicted image from the model.

We further evaluated the results using the mean Intersection Over Union Metric (mIoU) (Long, J 2015)

$$IoU_k = \frac{TP_k}{TP_k + FP_k + FN_k}$$

$$mIoU = \frac{1}{N}\sum_k IoU_k$$

This takes the number of correct pixels from each predicted image and compares that to the ratio of true positive pixels + false positive pixels + false negative pixels. This is done for each image in the training set. The final score is the mean of the individual results in the set. This gives a fine-tuned measure of the model's performance – evaluating its results on a pixel-by-pixel basis.

Based on (Dai 2022) a reasonable target for these two values on the AI4Mars dataset is between 0.7 and 0.8.

By ensuring the only change between the two approaches in the number of parameters in the model we can isolate the effects of the results to the model size alone.

Results and Interpretation

As seen in the results table both models obtained similar results on the Dice Coefficient and mIoU metrics. This suggests that the UNet was over-sized for the problem at hand. It may be the case that we can reduce the number of parameters for the model even further, but further comparisons are beyond the scope of this work.

It's also worth noting (although perhaps expected) that the smaller model took considerably less time to train – reducing training time from ~588 minutes to ~31 minutes. Evaluation of the data was also much faster, reducing from 1m 30s to 1m3.8. Its possible at this point that the system was more limited by the time it took to load and preprocess the images than the network itself.

Model	**Training Time**	**Testing Time**	**Dice Coefficient**	**mIoU**
UNet	588m 37s	1m 30s	0.86	0.76
Mini-UNet	31m 15.9s	1m 3.8s	0.86	0.76

Example output from each model can been in figures 4 and 5. As might be anticipated by their similar Dice Coefficient and mIoU their outputs are largely similar. Of note is the fact that both models missed the two large rocks in fifth row.

Figure 4 UNet Segmentation Sample

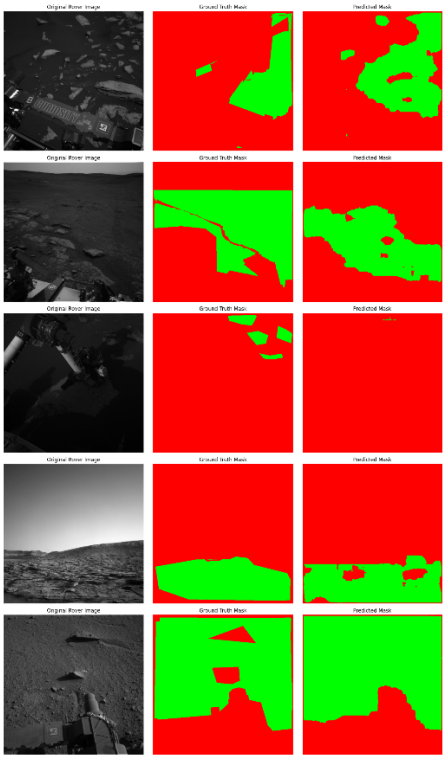
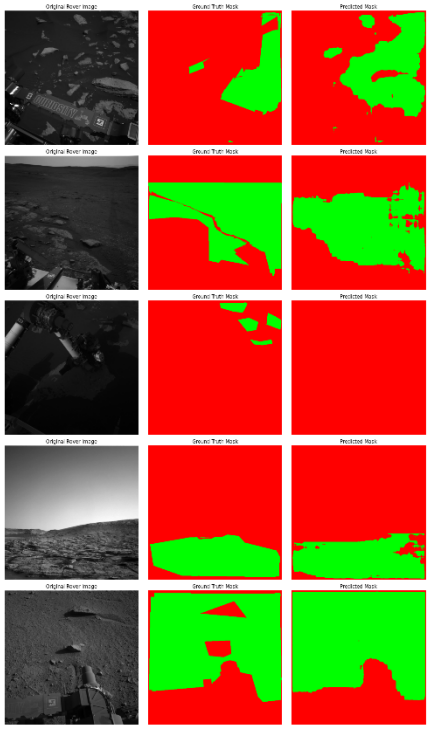


Figure 5 Mini-UNet Segmentation Example



Limitations and Risk

This paper investigated the effects of reducing the parameters on UNet model for solving a semantic segmentation task. While it turned out that on the available data, reducing the size of the network had no appreciable effect, we should keep the following points in mind:

We only trained the models on data from the Curiosity Rover. This means it's limited to the small subsection of Mars that the Curiosity Rover has visited.

As mentioned in the previous section, both models missed the large rocks in the bottom row. Missing these rocks in a real-Mars scenario could've resulted in damage to the Rover.

While we've tested the model on data collected by the Rover, we've never tested the model in a real-time scenario. It's possible that once deployed to a real Rover that we'd see unexpected results that could put the Rover in danger.

Ethical and Responsible Use

Given the cost of Mars missions, as well as the potential opportunity loss from a scientific point of view, use of this model without further refinement in a real-world Rover would be an unethical use of resources.

In addition, as mentioned in the section on risk, the training data is heavily based not only on the small area of Mars that Curiosity was able to reach, but potentially to any artifacts / idiosyncrasies in the Curiosity Rover's camera system itself. It's possible that if either of these models were used with data collected from a different camera setup, we'd get unreliable results.

The results of misusing this model could be the catastrophic loss of multi-million-dollar Rover. For example, if the Rover encountered the rocks shown in the final row of figures 4 & 5, the model would completely miss it, possibly resulting in the Rover colliding with the rocks.

Future Improvements

The next logical step for this work is to attempt to reduce the size of the model even further in a bid to see how small the network can be made before we see a reduction in performance. We could also look at the effects of data augmentation / feature engineering (such as enhancing images using traditional computer vision techniques before feeding them to the network) on performance.

Another step we can take is to integrate more data from the AI4Mars dataset. The AI4Mars dataset has data from Curiosity, Spirit, Opportunity, and Perseverance. This paper only made use of data from Curiosity Rover. Perhaps with more diverse data we could improve the performance of the model further.

References

Dai, Y., Zheng, T., Xue, C., & Zhou, L. (2022). SegMarsViT: Lightweight Mars Terrain Segmentation Network for Autonomous Driving in Planetary Exploration. *Remote Sensing*, *14*(24), 6297. https://doi.org/10.3390/rs14246297

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Swan, R. M., Kiri, L., Ono, M., Rothrock, B., & Higa, M. (2021). AI4MARS: A dataset for terrain-aware autonomous driving on Mars. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2021)*. https://openaccess.thecvf.com/content/CVPR2021W/AI4Space/html/Swan_AI4MARS_A_Dataset_for_Terrain-Aware_Autonomous_Driving_on_Mars_CVPRW_2021_paper.html

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